Supplementary Figure SB — every qualifying subsystem (≥5 reps in ≥1 cell)

E1 q<0.01 (dot between the two adjacent columns) · E2 q<0.05 (arrow between adjacent blocks in the regimen row) · Shaded rows in bold · Grey = subsystem absent from that cell

Beta oxidation of di-unsaturated fatty acids (n-6) (mitochondrial) (n=6)
Beta oxidation of even-chain fatty acids (mitochondrial) (n=23)
Carnitine shuttle (endoplasmic reticular) (n=21)
Fatty acid biosynthesis (unsaturated) (n=8)
Fatty acid elongation (even-chain) (n=6)
Glycerophospholipid metabolism (n=40)
Leukotriene metabolism (n=32)
Omega-3 fatty acid metabolism (n=15)
Omega-6 fatty acid metabolism (n=13)
Sphingolipid metabolism (n=25)
Acyl-CoA hydrolysis (n=54)
Beta oxidation of di-unsaturated fatty acids (n-6) (peroxisomal) (n=6)
Beta oxidation of odd-chain fatty acids (mitochondrial) (n=8)
Beta oxidation of unsaturated fatty acids (n-9) (mitochondrial) (n=5)
Beta oxidation of unsaturated fatty acids (n-9) (peroxisomal) (n=9)
Carnitine shuttle (mitochondrial) (n=25)
Eicosanoid metabolism (n=8)
Fatty acid activation (cytosolic) (n=65)
Fatty acid biosynthesis (n=24)
Glycosphingolipid biosynthesis-ganglio series (n=11)
Aminoacyl-tRNA biosynthesis (n=20)
Beta oxidation of even-chain fatty acids (peroxisomal) (n=12)
Fatty acid activation (endoplasmic reticular) (n=55)
Glycerolipid metabolism (n=29)
Acylglycerides metabolism (n=16)
Arachidonic acid metabolism (n=28)
Carnitine shuttle (cytosolic) (n=50)
Fatty acid biosynthesis (even-chain) (n=5)
Fatty acid desaturation (even-chain) (n=11)
Fatty acid oxidation (n=168)
Prostaglandin biosynthesis (n=7)
Cysteine and methionine metabolism (n=13)
Glycine, serine and threonine metabolism (n=27)
Tyrosine metabolism (n=19)
Valine, leucine, and isoleucine metabolism (n=28)
Arginine and proline metabolism (n=37)
Histidine metabolism (n=7)
Phenylalanine, tyrosine and tryptophan biosynthesis (n=48)
Protein assembly (n=17)
Alanine, aspartate and glutamate metabolism (n=20)
Lysine metabolism (n=10)
Protein degradation (n=6)
Tryptophan metabolism (n=5)
Beta-alanine metabolism (n=10)
Protein modification (n=5)
Androgen metabolism (n=7)
Cholesterol biosynthesis 1 (Bloch pathway) (n=5)
Estrogen metabolism (n=6)
Cholesterol metabolism (n=10)
Formation and hydrolysis of cholesterol esters (n=16)
Bile acid biosynthesis (n=26)
Steroid metabolism (n=21)
Amino sugar and nucleotide sugar metabolism (n=13)
Purine metabolism (n=51)
Pyrimidine metabolism (n=37)
Nucleotide metabolism (n=47)
Butanoate metabolism (n=5)
Chondroitin sulfate degradation (n=17)
Inositol phosphate metabolism (n=53)
Keratan sulfate degradation (n=7)
N-glycan metabolism (n=23)
Pentose and glucuronate interconversions (n=7)
Pentose phosphate pathway (n=20)
Blood group biosynthesis (n=6)
Glycolysis / Gluconeogenesis (n=20)
Propanoate metabolism (n=13)
Pyruvate metabolism (n=20)
Fructose and mannose metabolism (n=11)
Tricarboxylic acid cycle and glyoxylate/dicarboxylate metabolism (n=25)
Chondroitin / heparan sulfate biosynthesis (n=9)
Biopterin metabolism (n=5)
Nicotinate and nicotinamide metabolism (n=13)
Folate metabolism (n=16)
Glutathione metabolism (n=9)
Porphyrin metabolism (n=7)
Retinol metabolism (n=20)
ROS detoxification (n=7)
Drug metabolism (n=35)
Pantothenate and CoA biosynthesis (n=5)
Miscellaneous (n=9)
Transport reactions (n=739)
Exchange/demand reactions (n=141)
Pool reactions (n=5)

BL 24 | D 24 | LD 24 | BL 48 | D 48 | LD 48 | BL 72 | D 72 | LD 72 | BL 96 | D 96 | LD 96 | BL 120 | D 120 | LD 120

Cell (regime over hpf)

Row z-score of mean centered $\log_2|\text{flux}|$

3
2
1
0
−1
−2
−3